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Reading Science Articles

Learning how to write effectively about data begins with a strategy for reading and examining how the author organizes and writes about their findings. When we read to write, we identify the main components of an analysis and notice how the author brings these components together to form a logical and compelling story. During this process, we discover examples and templates that we can use to organize our own work and write about our findings. To get started, the following three steps can be helpful.

- *Identify the elements of the data analysis.* We begin by looking throughout the article for various building blocks of the data analysis. We often find that some of the elements of a data analysis are included in the article, while others are not. This investigation helps us understand the choices that an author makes in writing about their findings, and, as a reader, we assess whether particular omissions impact the credibility of the conclusions or whether any included details are superfluous to the main story.
- *Examine the argument.* When we read an article, we expect to be convinced of the importance and validity of the findings. We look for context that explains how the findings fit with others' work, and we try to discern whether the new insights support, counter, or extend current views in the field. We consider the appropriateness of the analysis, the generalizability of the conclusions drawn, and whether others' work have been adequately and convincingly presented. To help make this assessment, we pay close attention to the words the writer uses and consider how an alternative, similar word choice could impact the strength of a claim. We also pay attention to the article's tone and whether it adds credibility and to the connections between paragraphs and between sections and whether they convey a cohesive story.
- *Map the organization of the document.* At a basic level, science articles have three main parts—a beginning, which serves to define and motivate the problem; a

middle that presents the findings and explains what they mean; and an end, which summarizes the conclusions and their importance. Mapping out these parts of the document helps us see how the author chooses to organize their analysis, gives a sense of what a reader might expect when reading an article, and provides templates that we might follow.

The remaining sections of this chapter describe how to carry out these steps for reading an article. We place ourselves in the position of a reader and identify what we look for when we read with an eye towards writing. Several examples are provided; some come from publications aimed at a broad audience, such as a popular statistics magazine and a journal focused on the history and culture of mathematics. Other examples are from more technical science journals but do not require a lot of background in the field to understand the core argument. Most of these articles are about four pages long so you can quickly read them in their entirety. Links to them can be found in the notes section at the end of the chapter.

This description of how to read an article also serves to introduce the goals of the book, and it provides a map of the topics covered in later chapters. Our main focus is on writing a technical article, but we also address how to write for other audiences. If general science communication interests you, then Chapter 2 extends the approach for reading technical articles to press releases and blog posts. Additionally, writing for these other “publics” is covered in designated sections and activities in Chapters 6–10.

1.1 Identify the Elements of a Data Analysis

If we are to be convinced of the conclusions of a data analysis, we must have enough information to assess the quality of the data. We need to know about the kind of data collected and the conditions under which they were collected. This information allows us, as readers, to evaluate the appropriateness of the methods used to analyze the data, decide how reasonable the conclusions that the author draws are, and assess the limitations, generalizability, and impact of the study. We look for the following aspects of the data and their analysis.

Design The question of how the data arose is the foundation to understanding an analysis. The data at hand can be generated from an experimental or observational process. Or, the analysis can be model-based and rely on summary statistics, or a meta-analysis that collectively examines several earlier studies.

Protocol The conditions under which the data were collected are critical. We want to know about the setting of the study, such as the time period, location, etc., and how subjects were selected, e.g., a probability sample, self-selection, or an administrative collection. Additionally, we also want to read about the outcome (i.e., response) measurements, covariates, and about nonresponse, including missing values, drop-out,

and lost to follow-up. If appropriate, we also want information about treatments used, method of treatment assignment, and blindness of subjects and evaluators.

The description should be thorough enough to assure us that the analysis and conclusions are appropriate, and we expect the author to mention any limitations encountered in the data collection that might affect the conclusions. The amount of detail provided can vary a lot. For example, when the source is reliable and well-documented, such as a government survey, then we are satisfied if the author simply refers us to other, easily available documentation. Chapter 3 covers the topic of writing data descriptions.

Data Analysis There is an abundance of statistical methods available to analyze data, and our aim when reading to write is to observe how an author supports their findings through the presentation of specific results from the analysis. An analysis can be organized loosely into three categories: summary statistics and tables, statistical graphs, and formal methods, such as confidence intervals, hypothesis tests, and regression models.

A statistical graph is often used to convey the key finding of an analysis. Graphs are also provided as alternatives to tables of summary statistics and to support decisions made in choosing and assessing formal methods. Guidelines for making effective statistical graphs are provided in Chapter 4.

In addition to writing an article, the author often provides the code they wrote to clean and analyze the data. Writing code is another form of communication, and science writers are increasingly expected to make their code available for others to read, evaluate, and apply. A few simple guidelines for writing code to make it accessible and reproducible are provided in Chapter 5.

Complex statistical analyses often consist of a lot of assumption checking and modeling output, and choosing what information to provide and what form to provide it in is a crucial task for a science writer. For example, even when more advanced statistical methods are performed, summary statistics are often included to lay the groundwork for the analysis. Decisions about which technical material to include in an article is the topic of Chapter 6.

1.1.1 Determining if the Golden State Warriors Have Hot Hands

The *Mathematical Intelligencer* publishes articles written in an engaging, informal style aimed at what the journal calls a “general intellectual audience”. For example in “Do the Golden State Warriors Have Hot Hands?” by Daks, Desai, and Goldberg, the researchers study the performance of star basketball players to assess evidence of a hot hand, i.e., whether a player is more likely to make a shot after having just made some shots. We consider how the authors describe the statistical elements of their argument for this broad audience.

At its core, the analysis consists of a hypothesis test. The authors gradually build to the formal definition of the test, starting with conceptual descriptions and examples. They first describe the test statistic in non-technical terms:

...the probability of making a shot given a string of two hits minus the probability of making a shot given an equally long string of misses.

Then to help the reader understand this definition, the authors compute the statistic for a concrete situation: Klay Thompson's shooting record in the December 23, 2016, game against the Detroit Pistons.

The record is represented by a string of 1's (hits) and 0's (misses):

1 1 1 0 1 0 0 1 1 0 0 0 0 1 1.

Thompson took 16 shots in this game, and as it happened, he made exactly half of them. ... There are four instances of two hits in a row, indicated by the string 11. We know what happened after the first three instances: Thompson hit the first time and missed the second and third times. But nothing happened after the fourth instance of 11 because the game ended before he could take another shot. ...we observed Thompson scoring one-third of the time following two hits in a row. In the other direction, given that Thompson missed twice in a row, he scored two-fifths of the time. ...The difference of the two conditional probabilities is $1/3 - 2/5 = -1/15$.

After the conceptual definition and empirical calculation, the formal definition of the test statistic is provided and notation defined.

The underlying assumptions of the hypothesis test are essential; these assumptions lead to the probability model for generating data under the null hypothesis and calculating a p -value. The authors put forth a general explanation that the null distribution of the test statistic comes from "a random string of hits and misses". They further explain that the test is conditional on a given string of hits and misses for a specific player in a particular game, i.e., the null distribution is defined by all possible permutations of the observed string of hits and misses. The null distribution and p -value are estimated via simulation. A histogram of the null distribution is provided for Thompson's string of shots in the December 5, 2016, game against the Indiana Pacers. The non-normal shape displayed in the figure helps us understand the issues that arise with short sequences, which the authors aim to address with their methodology.

Additional statistical elements in the article consider alternatives to this hypothesis test. The authors calculate p -values with and without a strict inequality; drop free throws from the sequence of hits and misses to see if that changes the findings; use other models of randomness, such as generating hits and misses from a coin toss where the chance of a hit matches the player's record; and compute alternative test statistics. Few of the specific results from these variants of the hypothesis test are included in the article. Instead, the authors provide the reasons for considering the alternatives, non-technical descriptions of them, and a brief summary of the findings and how they support their conclusions. Given that the authors have explained the main test in detail,

we have enough information to understand how they carried out the other versions of the test, and we can trust their findings.

1.2 Examine the Argument

A common misconception about science writing is that scientists do not make arguments when they write because science must be impartial and objective. However, the aim of a scientific article is to make a convincing case for the findings, and when we read an article, we are mindful of how the author makes their case. We become convinced of the importance of the work if we can judge its relevance and generalizability, see how the work fits with other results in the field, and determine that the author has written an honest and accurate portrayal of the findings. If we are to accept the findings as compelling evidence, we expect the scientist to accurately describe them, place them in the context of other findings, and demonstrate a balanced understanding of their implications. The art of organizing your findings into a convincing story is the subject of Chapters 6 and 7. Below are a few ways we examine the argument as readers.

Placement The findings from a data analysis need to be presented in context. In order to understand the relevance of new findings, we, as readers, need to know how these new results fit into existing theories. These findings may: support existing understanding and possibly fill a hole in current theories; address disagreements in the literature by providing an argument for one theory/approach over another; or advance a new explanation or a new method for understanding the world, which may express a critique of existing perspectives.

Generalizability Data are crucial to scientific arguments, and the quality of the data and the analysis are key to assessing the implications of new findings. An author should identify what they can say about their findings and what they cannot. Addressing both the limitations and the generalizability of findings go hand-in-hand, and we expect to see both in a well-written article.

Relevance The author must convince the reader that the identified problem of interest will help advance knowledge in a field and that the approach is relevant to the problem of interest. For findings to be considered impactful, they must reflect careful and sound data analysis, and the conclusions must be appropriate for the analysis—neither overblown nor too cautious. In short, the problem must be relevant to the reader, and the solution must be relevant to the problem itself.

Language A science writer must take care with the wording they use to make their argument. With the wrong choice of a word, an argument can easily slip into a stronger claim than what is supported by the data and analysis. When we read, we pay close attention to the precise wording of the author's statements, and reciprocally, the author

must ensure that their writing accurately reflects the findings. Additionally, terms such as *significance* and *confidence* have special connotations in statistics so science writers should avoid them in their everyday usage. More information about careful use of statistical terms can be found in Chapter 8, and general advice for crafting sentences appears in Chapter 9.

1.2.1 Assessing a New Method to Weigh a Donkey

Case studies in *Significance* magazine offer examples of data analyses that are accessible to readers with differing levels of statistical training. These articles are generally engaging and relatively easy to read. They are written in a somewhat informal style, yet authors are careful in how they report findings and draw conclusions. As an example, we consider the article, “How to weigh a donkey in the Kenyan countryside” by Milner and Rougier. In this article, the authors describe a method they have developed to get a quick and accurate estimate of the weight of a donkey using only the measurements of the animal’s length and girth.

We examine the wording in the article’s two concluding paragraphs. The penultimate paragraph (below) presents the model and gives a concrete example of its application. We draw attention to three particular phrases: *Readers can confirm themselves*; *donkey number 78*; and *rounded to the nearest kilogram*.

We refitted the model with these recoded factors, taking the most populous levels of body condition and age as the reference ($BCS = 3$ and $age \geq 5$). Our resulting model is: ...Readers can confirm themselves that a donkey of more than 5 years of age, with a BCS of 3, a girth of 122 cm and a length of 103 cm, weighs approximately 175 kg. This corresponds to donkey number 78 in our dataset, whose actual weight is 183 kg. The relative error is about +5%. The additive adjustments for those donkeys of different ages and body condition are given in Table 1, rounded to the nearest kilogram.

When the authors write *Readers can confirm themselves*, they informally invite us to work out an example to check our understanding of the method. To explain the assessment of the method in simple terms, the authors chose a precise set of input values that correspond to a donkey in the dataset, which they refer to as *number 78*. Of course, we don’t really need to know the number of the donkey in the dataset. It is simply a device to engage the reader. To avoid confusion, the authors explain that the values in Table 2 are *rounded*.

The final paragraph below summarizes the accuracy of the new method using data that were set aside (not included in the model building exercise). Note the careful qualifications made by the authors with the use of the phrases *proceeds exactly*, *reasonable*, *typical*, *about*, *relatively*, and *over the range*.

Finally, we assessed our tool’s accuracy, using the hold-out sample of one-fifth of the donkeys that we mentioned previously. The prediction of weight proceeds

exactly as if we were in the field; that is, we used only the information in Figure 4 and Table 1 to obtain our predictions, which we could then compare to the donkey's actual weights. Figure 5 and Table 2 show that it is reasonable to claim that the typical accuracy of our tool is about $\pm 10\%$, and that this is relatively consistent over the range of predicted weights from 75 to 200 kg.

The authors explain that the assessment is fair because it *proceeds exactly* as in the field. They summarize the accuracy by examining *typical* values; use rounded figures for simplicity (*about $\pm 10\%$*); and evaluate the accuracy for donkeys similar to those for which they have data (*over the range*). Note also that their statement of results are softened by calling them *reasonable* and *relatively* consistent. Consider a sentence without these qualifiers: “We claim that the accuracy of our tool is 10%.” Such a claim might leave the authors open to criticism when, e.g., the method is applied to a 50-kg donkey which is outside the observed range.

1.2.2 Helping the Scientific Community Adopt a New Standard

The article “Evaluation of the accuracy, consistency, and stability of measurements of the Planck constant used in the redefinition of the international system of units” by Possolo et al. helped usher in a new definition of the kilogram. This article was published in *Metrologia*, which according to the journal's website aims to “publish papers that contribute to the solution of difficult measurement problems and improve the accuracy of derived units and constants that are of fundamental importance to physics”.

For the scientific community to redefine the kilogram, a consensus needed to be reached, and scientists needed to be convinced that the community was ready for the change. The Possolo study advocates for readiness to adopt the new definition. The investigators carry out a meta-analysis where they construct an argument from an evaluation of other studies. That is, their work does not propose a new method or present new data; instead, the primary contribution is a comprehensive and accurate portrayal of the work of others.

The introduction to the article lays out the main threads of their argument: assessment of the accuracy of the measurement of the Planck constant; consistency of the measurement across laboratories; stability of the measurement process over time; and a comparison of the reliability of the new kilogram measurement to the physical standard. The authors take these issues in turn and dedicate a section of the paper to address each.

The first avenue the authors consider is the question of stability. Measurements of the Planck constant have been made since 1979 at national laboratories across the world. These laboratories publish their measurements and their uncertainty, and the authors examine results from the various studies. A figure in the paper that displays a plot of the estimate (and its uncertainty) over time is particularly convincing for the redefinition because the measurements made over the past several years appear stable and accurate.

The accuracy and consistency results described in the article are situated on the world metrology stage. To lay the groundwork, these sections of the article each begin with a

statement from the Consultative Committee for Mass of the International Committee Weights and Measures (CCM) on the prerequisites needed for adopting a new definition. For example, the accuracy section of the article quotes the CCM recommendation: “at least three independent experiments, including work from Kibble balance and XRCd experiments, yield values of the Planck constant with relative standard uncertainties not larger than 5×10^{-8} .” The authors argue that six experiments have achieved the CCM threshold.

The consistency section includes a critique of the commonly used approach to assess consistency. The authors point out the flaws of the “standard” statistic, consider an alternative, more-appropriate statistic from recent literature, and make the case that even with the flawed assessment the size of the inconsistency is negligible in comparison to accuracy.

Finally, the conclusions section of the paper makes several strong statements in favor of readiness for redefinition. This section first addresses reliability:

The level of agreement between the most accurate determinations of h available today is simply remarkable, especially considering that they involve multiple, different, and independent versions of two radically different approaches to its measurement.

The authors describe the accomplishment in glowing terms:

...a feat that has not been attempted since the national convention of the First Republic of France passed a decree on April 7, 1795, defining the gram as the 'absolute weight of a volume of pure water equal to the cube of one hundredth part of a meter, at the temperature of melting ice'.

The conclusions bring two additional points into the argument: there is a benefit to moving to this new definition with respect to the time and cost of dissemination; and continued research can improve the measurement techniques and further reduce uncertainty.

1.3 Map the Organization

A science article that reports on findings from a data analysis typically contains standard elements for presenting findings and making an argument, e.g., background information, methods description, summary of findings, and discussion of results. When reading to write, we note where in the article these elements appear.

When we read, we also look to the section headings to guide us and let us know what to expect and to reveal the organization of the story. When we notice placements of different types of findings, we can get ideas for how to organize our own story. Likewise, we want the wording of the title to accurately reflect the findings of the study.

Below is an overview of what to look for and where. More information on how to organize a technical article can be found in Chapter 7.

Context In an article on a statistical analysis, the author provides information about the data and the protocol used in collecting the data so that we, as readers, can use this information to assess the validity and importance of the findings. More specifically, we look for a description of the data (see Chapter 3) and how and why the data were collected. This description typically appears soon after the introduction to lay the groundwork for framing the analysis. Additionally, a literature review (see Section 7.5.1) places the current study in the context of related earlier findings. Review material often appears at the beginning or early in the middle part of the article. However, there are times when this information is embedded in a discussion (see Section 7.4.1) of the findings at the end of the article.

Methods The statistical methods presented in a report range from exploratory data analysis (EDA) to more formal analyses (see Section 7.3). EDA is typically discussed prior to presenting the formal analysis, and sometimes the findings from EDA are mentioned only briefly to confirm that the data are as expected. Other times EDA serves to motivate the approach taken in the formal analysis, and sometimes the article contains no formal analysis and consists entirely of EDA. A formal analysis might include an assessment of the effectiveness of a model, a comparison of findings to those that use a different, competing method, or a summary of estimator accuracy. These assessments and comparisons offer context that help quantify the impact and typically they come near the end of the presentation of the methods or in a separate discussion.

Statistical Graphs and Figures Visual displays can be useful in supporting study findings; they can justify a choice of method used or assess the impact of the analysis. Often a statistical graph, along with its caption (see Section 7.3.5), can make a compelling case for us to understand and judge the findings, and when such a central graph is placed early in the article it can help us readily grasp the main results without getting into the details of the analysis. See Chapter 4 for how to make and read graphs.

Impact Explanations of the impact of the analysis often appear in many places in an article. For example, the abstract (see Section 7.5.3) and introductory material (see Section 7.5.2) usually include a high-level description of the findings, the explanation of the analysis in the middle of the article often contains a more-detailed description of the impact of the findings and intermediate results, and the end of the article typically gives a high-level summary of the findings, their implications, and relationship to other work.

1.3.1 Organizing the Announcement of an Exoplanet

Astronomy & Astrophysics is an open-access journal that publishes articles in theoretical, observational, and instrumental areas of astronomy and astrophysics. In the article

“Transiting planet candidate from K2 with the longest period”, Giles et al. describe new methodology developed to discover planets that have long orbital periods, and they apply these methods to data collected from the space telescope K2 to find a new exoplanet (a planet in another solar system).

The diagram in Figure 1.1 shows the layout and section headings of this four-page article. While this article is more technical than others in this chapter, we focus primarily on the paper’s organization. The headings and their sequence in the paper follow general guidelines and reveal little about the specific content (introduction, observations, analysis, discussion, and conclusions). The abstract also includes standard identifiers: context, aims, methods, results, and conclusions.

The first paragraph of the introduction provides motivation for the new method and includes a brief literature review for context. A detailed description of the data appears in the observations section. Two figures are included in this section. The first of these figures provides compelling insight into the new methodology. This central graph shows an unusual dip in the light curve of a star, which the authors explain is how the transit of a star might be detected from K2 data. The second plot confirms that the structure observed in the first plot corresponds to a planet; this plot is based on follow-up data collected from the Euler telescope in Chile.

The diagram in Figure 1.1 shows that the formal modeling used to estimate stellar and planetary parameters occupies at least half the space in the article.

The discussion at the end of the paper focuses on the impact of the research. This section includes the caveat that it will take three years to confirm results. The relevance of the findings are conveyed in the claim that this is the longest orbital planet discovered to date, and in likening the planet to the gas giants in our solar system.

1.4 A Guide to Reading as a Writer

This chapter aims to prepare you to write by first helping you read from the perspective of a science writer. We chronicle how a science writer describes the analysis that forms the core of their scientific finding.

We don’t expect you to read an article straight through from beginning to end. When we read science articles, we typically make several passes over the article, each time understanding the material in greater detail. One common approach is to begin by reading the abstract and introduction and then going on to examine the figures and tables and reading their captions. This first pass gives a high-level view of the results, and from there we might choose to read the discussion to understand the impact, the data description to assess whether the strength of the claims is justified, or the methods for more details on the proposed methodology. Eventually we read the entire article through from start to finish without jumping around. This complete read is especially useful to get a sense of how the author makes their argument.

In this section we provide guidelines to follow while reading that are parallel to the three aspects of reading a statistical report described in Sections 1.1, 1.1.1, and 1.3.

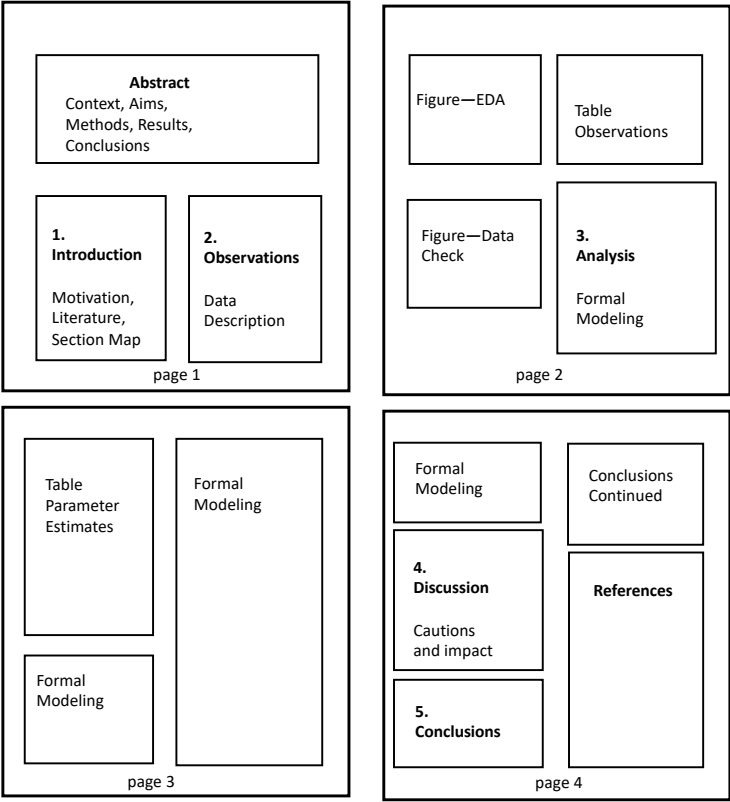


Figure 1.1 Layout of a Four-Page Article. This diagram shows the layout of an article on the discovery of an exoplanet. This organization is typical of a technical article published in a science journal.

1. *Statistical Elements.* A starting point for reading to write is to capture the analysis in a few broad strokes. The following tasks will help you in this process.
 - a. *Summary.* Write a one- or two- sentence summary of the objective of the study.
 - b. *Data.* Describe the data used in the study by following the guidelines provided in Section 1.1. If there are any aspects of the data collection and design not captured by these guidelines, describe them.
 - c. *Analysis.* Briefly summarize the statistical analysis and the main outcomes. If possible, note any details excluded from the article.
 - d. *Conclusions.* Write a short summary of the conclusions drawn from the data analysis.
2. *Argument.* Consider the following questions about how the writer positions their findings and their use of language.
 - a. *Placement.* How do the results presented relate to the rest of the scientific literature? Do they support existing understanding of a phenomenon, fill in the gaps in current understanding, offer evidence in a debate, or provide a new approach or new evidence to study a phenomenon?
 - b. *Generalization.* Are there limitations or difficulties in generalizing beyond the data? Are these limitations noted? Do you have any other concerns about the study?
 - c. *Impact.* How does the study advance knowledge in the field? Do the author's claims seem adequately justified, overblown, or unduly cautious?
 - d. *Language.* How do the specific words the author uses strengthen, clarify, and support their argument? Consider the following in answering this question.
 - Identify a paragraph that summarizes the result of the analysis. Examine the choice of words used, and relate them to the strength of the conclusions and the limitations the author places on the generalizability of the findings.
 - If you could change one sentence so that it is clearer, more concise, or more precise, which sentence would you choose? How would you change it, and what are your reasons?
 - How accurately and completely does the title of the article reflect the study findings?
3. *Organization.* Obtain a high-level picture of the organization by making a map of the article layout like the one in Figure 1.1. Mark the map with locations for the: objective and motivation; type of study and study protocol; description of methods and findings; statistical analysis; graphs and tables; related findings

made by others; discussion of limitations, implications, and generalizations; and conclusions. On the map, denote which pieces of the analysis are exploratory, formal, or a comparison/assessment.

Use the map to describe in a few sentences where the elements of the statistical analysis and the argument are found in the article. Specifically, address the following issues.

- a. *Context.* What context is provided? Are related findings used to motivate the topic or are they provided in a later discussion?
- b. *Methods.* How much space is dedicated to each type of analysis, i.e., exploratory, formal, comparison and assessment? Consider the level of detail in the method sections. What aspects of the analysis are mentioned without detail given, e.g., alternative models, assumption checking, confirmation of others work. How does the level of detail correlate to the background of the intended audience?
- c. *Graphs and tables.* How are the graphs and figures used to support the findings? Do they address EDA, formal analyses, or comparison and assessment? Would additional figures be helpful? If so, what kind? Identify one key graph. Where is it located in the paper? What message does it convey with respect to the objective and conclusion of the study?
- d. *Impact.* How often is the impact of the study discussed? Does the discussion of impact appear in the abstract, introduction, middle, and/or conclusions? Compare the level of detail in addressing the impact in these various sections of the article.

1.5 How to Weigh a Donkey in the Kenyan Countryside

As an example of how to use the guidelines in Section 1.4, we read and take notes on the article “How to weigh a donkey in the Kenyan countryside”, by Milner and Rougier.

1. Identify the Statistical Elements

a. *One-sentence summary.* The aim of the study is to develop a simple tool to estimate the weight of a donkey from easily obtained measurements such as length and girth so that veterinarians in the field can prescribe medicine to sick animals without needing to weigh the animal directly.

b. *Data.* The data came from an observational study of donkeys. The study took place in July–August 2010 at 17 sites located in the Yaha and Naivash districts of rural Kenya. The participants were 544 donkeys from 2 species that were brought to the Donkey Sanctuary for de-worming. In some locations all donkeys were included in the study; in other locations a sample was taken. Pregnant donkeys and donkeys with visible disease were not entered into the study. Donkeys were measured by those working with the

Donkey Sanctuary. It is not stated whether there were multiple recorders. Recorders judged the condition of the donkey and scored them on a scale from 1 (emaciated) to 3 (healthy) to 5 (obese). The outcome is the weight (kg) of the donkey, and the covariates are girth (cm), height (cm), length (cm), sex (three categories: stallion, gelding, female), age (six categories), and body condition (potentially nine categories, but only six were observed). Three donkeys were excluded: one was too young, one was emaciated, and the third was obese. The results of the study were generalized to all donkeys in rural Kenya. Since no treatments were given, the question of blindness is not applicable. However, 31 donkeys were weighed twice to check the accuracy of the measurements. Those donkeys were unmarked after the first measurement so that the recorder wasn't aware of making a second measurement on a donkey. No information is provided about how donkeys were selected when too many arrived at the deworming station (e.g., whether a probability sample was used).

c. Statistical Analysis. The data were divided at random into a test set (one-fifth of the donkeys) and train set (the remaining four-fifths). The train set was used to select the model, and the test set was used to evaluate the model. The Box—Cox transformation was used to select a transformation for the outcome variable. The main statistical method was multivariate linear regression, which included categorical variables and used a modification of squared error to adjust for a different penalty for under or over estimating the weight. A small set of models were compared. The final product was a parallel-scale nomogram—a visual tool that converts combinations of two input values into an output value.

The article contains three figures and two tables. One figure shows a Box—Cox plot of loss to explain the choice of transformation for weight. A scatter plot of the size of the effect for the categorical variables justifies the collapse of some categories. The last figure contains a scatter plot of actual versus predicted weight for the test set to help visualize the relative error. The first of the two tables contains values for the size of the effect for the categories of body condition and age, and the second table gives the relative error in the test set.

In addition to the statistical graphs and tables, the article contains several other figures and diagrams, which add interest and help convey how the new method works. These visual displays are a photograph of two donkeys, a diagram of how to use a nomogram, a drawing of two loss functions for different medications, and a figure of the final nomogram constructed from the fitted model.

d. Conclusions. The model developed has a 10% error rate for donkeys with predicted weights in the 75–200 kg range.

2. Examine the Argument

a. Placement. The results are well situated in the scientific literature. The article includes a discussion about more complex nomograms and why they would be problematic in this case, and evidence that the nomogram method is in use for other kinds of animals. This study offers a new approach that maintains the simplicity of the two-variable nomogram, but offers more complexity by including additive effects.

b. Generalization. Generalizing to donkeys in rural Kenya seems reasonable. One caution: the donkeys in the study were brought to the Donkey Sanctuary to be dewormed, so the health of these donkeys might be different than the health of the typical donkey in rural Kenya.

c. Impact. The new tool advances the nomogram technique in a simple, novel, and useful way. The findings have the potential to greatly improve veterinary practice in rural areas.

d. Language. The title intrigues us by promising a solution to a logistical question. The second-to-last paragraph explains the results, and the final paragraph gives more information about assessing the accuracy of the approach. The wording in these two paragraphs is discussed in Section 1.2.1.

We correct the ambiguous language of the following two sentences: “This loss function is the cost of getting it wrong. It depends, among other things, on the drug that is being prescribed.” The rewrite attempts to clarify what “it” means the two times it is used: “The cost of getting the dosage wrong is measured by the loss function. The cost depends, among other things, on the drug that is being prescribed.”

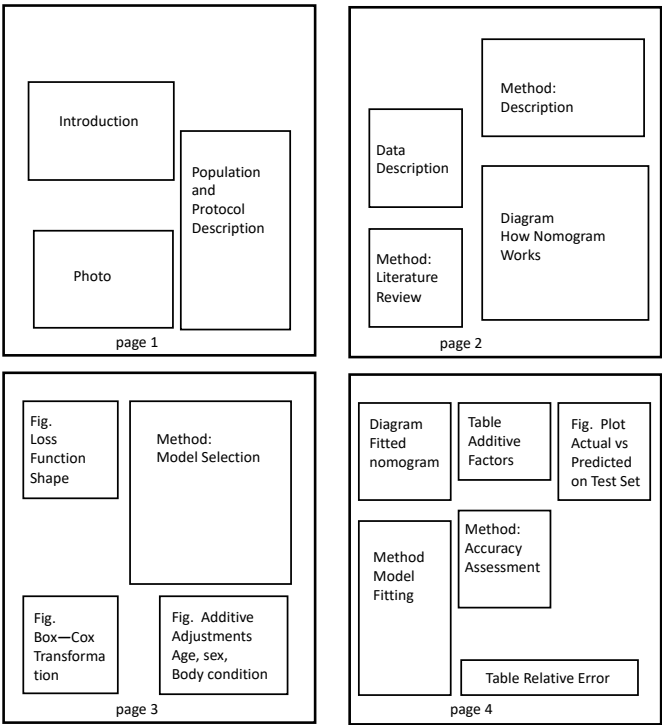


Figure 1.2 Layout of an Article on Weighing Donkeys. This diagram shows the article contains many visual components—figures, diagrams, and photo—to help convey the story to a broad audience.

3. Map the Organization The diagram in Figure 1.2 shows the layout of this four-page article. It locates the objective, study description, methods and analysis, figures and tables, literature review, discussion and conclusions.

a. Context. The topic is motivated in the introduction by a practical need of veterinarians in rural Africa. Additionally, after the description of the data, the notion of a nomogram and related work are provided. These references give motivation for the authors' novel statistical method.

b. Methods. More than half of the article is dedicated to the analysis, and about half of this space contains the formal analysis. The descriptions of all the methods are kept at a high level and little notation is used.

c. Graphs and tables. The core ideas, results, and assessment are explained with many diagrams and figures, which makes the material more broadly accessible. One key graph, a scatter plot of actual versus predicted weight for the test set appears at the end of the article and offers convincing evidence of the effectiveness of the method. A second key graph motivates the use of additive adjustments, which is the main insight of the authors' new method. This plot appears earlier in the article, where the method is described and the model fitted.

d. Impact. The impact of the study is discussed in three parts of the article. The discussion of impact first appears in the introduction as a what-if scenario, and then in the background literature where the new approach is contrasted to existing methods. Finally, the conclusion of the article discusses assessment.

1.6 Notes

The idea of learning statistics by reading news stories and science articles that report on statistical findings is described in Gelman and Nolan [5], and the guidelines in Section 1.4 are an adaption and extension of templates found there. Here the focus is on learning to write by reading technical articles from the perspective of a writer.

For more on how to examine arguments, see Graff and Birkenstein [9] and, in particular, the chapter called "The data suggest: writing in the sciences" by Gillen [7].

The articles used as examples in this chapter are openly available on the Web. More specifically, the article on hot hands in basketball by Daks, Desai, and Goldberg. [3] appears in the *Mathematical Intelligencer* (an earlier version [2] is available on arXiv). The article on the Planck constant by Possolo et al. [15] appears in *Metrologia*. Also, *The New York Times* provides a more general overview of the science and the people involved in changing the kilogram's definition (see Lim [12]). The article by Milner and Rougier [14] on how to weigh a donkey appears in *Significance*. The article by Giles et al. [6] on detecting planets orbiting other stars appears in *Astronomy & Astrophysics*. Additionally, a blog by Jorgenson [10] and press release [17] on the topic provide less technical descriptions about the new method for discovering exoplanets.

For more practice with reading to write, consider working on Portfolio Piece C.

1.7 References

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1.8 Activities

1.8.1 Reading to Write

The case studies in *Significance* magazine offer examples of data analyses that are accessible to readers with a basic level of statistical training. These articles are generally engaging and relatively easy to read. One possible drawback with using them to study how to write is that they are written in a somewhat informal style. If this style of writing is different from the style of the journals and magazines for which you want to write, then you will probably want to select articles from other publications for this activity.

The following short articles are on a variety of topics—nature, economics, sports, public policy, and history. Choose one that interests you and write a summary that follows the guidelines in Section 1.4. Section 1.5 provides an example of how to do this. “Neonicotinoids and bees: What’s all the buzz?” (Goulson, D. [8]).

“Do what the neighbors do: Reopening businesses after Hurricane Katrina” (LeSage, J. et al. [11]).

“Beyond reasonable doubt: catching the drug cheats at the London Olympics” (Erotokritou-Mulligan, I. et al. [4]).

“August’s child is... favoured by fortune” (Brewer, L. M. and Cochran, J. J. [1]).

“Alcohol, babies, and the death penalty” (Mardia, K. et al. [13]).

“The hazards of slavery” (Smith, S. and Forster, M. [16]).

1.8.2 Sharing Article Summaries

Write a one-paragraph synopsis of the article that you read for Activity 1.8.1 or for some other article of interest. This paragraph should encapsulate the goal of the analysis, the findings, and key features of the study. Ask someone who has not read this article to read your paragraph and discuss the study with you. (Or exchange summary paragraphs with someone who has read a different article and take turns discussing your articles.) Determine whether the reader understands the main points of the analysis, and identify what aspects they were unclear about.

As you discuss your synopsis, be sure to address the following questions.

- What did your reader misunderstand about the article from reading your synopsis? How could you change your summary to help them better understand?
- If you exchanged paragraphs, consider what you and your partner’s articles have in common. How are they different? Consider the following questions.
 - Content. What types of graphs and statistical methods appear?
 - Detail. Does the author oversell or undersell the conclusions? How accessible do you find the description of the methods, considering the intended audience?
 - Organization. Where are the methods, conclusions, and discussion of potential flaws found?

Finally, rewrite your summary paragraph to address the inadequacies that were uncovered in your discussion.

1.8.3 Skipping Around as We Read

As mentioned in Section 1.4, at first we often do not read an article straight through from beginning to end. Instead, we make several passes over the article, each time understanding it in greater detail. In this exercise, you will get a sense of how much you can glean from a quick read of an article.

Choose an article from a journal in your field that has at least three statistical graphs. Read the abstract, introduction, and figures and their captions. Based on this limited knowledge of the paper, describe how each figure is used to help the authors make their argument. Select one figure that you think is core to the article. Explain your choice.